SIMATS ENGINEERING
ASSESSMENT TOOL 2

NS AVESH HUSSAIN
192525241

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : Case Study (Medium)
Assessment Tool Weightage: 30%

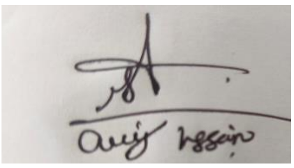
QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	**5**
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	**5**
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	**5**
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	**5**
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	**5**

6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	**5**
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	**5**
8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	**5**
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	**5**
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	**5**

Code of Conduct:

I NS AVESH HUSSAIN(192525241) ____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1. Online Shopping Platform – Normalize to 3NF
Relation:
ORDER(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)
Functional Dependencies

- OrderID → CustID
- CustID → CustName
- ProdID → ProdName, Price
- (OrderID, ProdID) → Qty

Candidate Key
(OrderID, ProdID)
3NF Decomposition
CUSTOMER
Customer(CustID, CustName)
ORDER
Order(OrderID, CustID)
PRODUCT
Product(ProdID, ProdName, Price)
ORDER_ITEM
OrderItem(OrderID, ProdID, Qty)
Result
The relation is decomposed into 3NF, removing partial and transitive dependencies and reducing red

2. Hospital – Normalize to BCNF
Relation:
PATIENT(PID, PName, DocID, DocName, Specialization, WardNo, WardName)
Functional Dependencies

- PID → PName, DocID, WardNo
- DocID → DocName, Specialization
- WardNo → WardName

Anomalies

Insertion anomaly:
A new doctor or ward cannot be stored without a patient.
Deletion anomaly:
Deleting the last patient of a doctor may delete the doctor's information.
Update anomaly:
Changing a doctor's specialization requires updating multiple patient records.
BCNF Decomposition
PATIENT
Patient(PID, DocID, WardNo)
DOCTOR
Doctor(DocID, DocName, Specialization)
WARD
Ward(WardNo, WardName)

3. University Enrollment – 2NF
Relation:
ENROLLMENT(SID, SName, CourseID, CourseName, Faculty, Grade)
Functional Dependencies
- **SID → SName**
- **CourseID → CourseName, Faculty**
- **(SID, CourseID) → Grade**

Candidate Key
(SID, CourseID)
Partial Dependencies
SID → SName
CourseID → CourseName, Faculty
These are partial dependencies because they depend only on part of the composite key.
2NF Decomposition
STUDENT
Student(SID, SName)
COURSE
Course(CourseID, CourseName, Faculty)
ENROLLMENT
Enrollment(SID, CourseID, Grade)

4. Airline Booking – Normalize to BCNF
Relation:
BOOKING(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)
Functional Dependencies
- **PassID → PassName**
- **FlightNo → DeptCity, ArrCity**
- **(FlightNo, PassID) → Seat, Date**

Candidate Key
(FlightNo, PassID)
BCNF Decomposition
PASSENGER
Passenger(PassID, PassName)
FLIGHT
Flight(FlightNo, DeptCity, ArrCity)
BOOKING
Booking(FlightNo, PassID, Seat, Date)

5. HR System – Normalize to 3NF
Functional Dependencies
Consider:
EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)
Dependencies:

- **EmpID → EmpName, DeptID**
- **DeptID → DeptName, ManagerID**

Therefore:
EmpID → DeptName, ManagerID
This is a transitive dependency.
3NF Decomposition
EMPLOYEE
Employee(EmpID, EmpName, DeptID)
DEPARTMENT
Department(DeptID, DeptName, ManagerID)
Result
The transitive dependency is removed and the database is normalized to 3NF. The document asks the HR case to identify functional and transitive dependencies and normalize to 3NF.

6. Library – 4NF
Relation:
LIBRARY(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)
Multi-Valued Dependencies
Assuming a member can independently have multiple books and the book details are independent:

- **MemberID →→ BookID**
- **MemberID →→ Title, Author, Genre**

Decomposition
MEMBER
Member(MemberID, MemberName)
BOOK
Book(BookID, Title, Author, Genre)
BORROW
Borrow(MemberID, BookID, BorrowDate)
Result
Independent multi-valued facts are separated, reducing redundancy and producing a 4NF design.

7. Telecom Billing – Keys and FDs
Consider a telecom relation:
BILLING(CustomerID, PhoneNo, PlanID, PlanName, CallID, CallDuration, Charge)
Functional Dependencies

- **CustomerID → PhoneNo**
- **PlanID → PlanName**
- **CallID → CallDuration, Charge**
- **(CustomerID, CallID) → all required call information**

Candidate Key
(CustomerID, CallID)
Primary Key
(CustomerID, CallID)
Other Keys
PhoneNo may uniquely identify a customer if the business rule guarantees one phone number per customer.

Result
Identifying keys and FDs before normalization helps remove redundancy and maintain data integrity. The question specifically asks for candidate keys, primary keys and FDs before normalization.

8. Retail Inventory – Complete Normalization
Relation:
PRODUCT(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)
Functional Dependencies

- PID → PName, SupplierID, Category, Price, Warehouse
- SupplierID → SupplierName

Transitive Dependency
PID → SupplierID → SupplierName
Decomposition
PRODUCT
Product(PID, PName, SupplierID, Category, Price, Warehouse)
SUPPLIER
Supplier(SupplierID, SupplierName)
Result
The supplier information is separated from product information. This removes the transitive dependency and reduces redundancy.

9. School ERP – BCNF Design
A poorly normalized School ERP may contain:
STUDENT(StudentID, StudentName, ClassID, ClassName, TeacherID, TeacherName)
Functional Dependencies

- StudentID → StudentName, ClassID
- ClassID → ClassName, TeacherID
- TeacherID → TeacherName

Transitive Dependencies
StudentID → ClassID → ClassName
StudentID → ClassID → TeacherID → TeacherName
BCNF Decomposition
STUDENT
Student(StudentID, StudentName, ClassID)
CLASS
Class(ClassID, ClassName, TeacherID)
TEACHER
Teacher(TeacherID, TeacherName)
Justification
Each determinant in the decomposed relations is a candidate key. Therefore, the relations satisfy **BCNF**.
Result
The School ERP is fully normalized, redundancy is reduced, and insertion, deletion and update anomalies are minimized.

10. Movie Streaming Platform – 5NF

A movie streaming system can contain independent relationships between:

Movie
Actor
Platform

Suppose:

STREAM(MovieID, ActorID, PlatformID)

Join Dependency

The relationship may be represented using:

MovieActor(MovieID, ActorID)

MoviePlatform(MovieID, PlatformID)

ActorPlatform(ActorID, PlatformID)

5NF Decomposition

MOVIE_ACTOR

MovieActor(MovieID, ActorID)

MOVIE_PLATFORM

MoviePlatform(MovieID, PlatformID)

ACTOR_PLATFORM

ActorPlatform(ActorID, PlatformID)

Lossless Join

The original relation can be reconstructed through joins of the decomposed relations according to the specified join dependency.

Therefore:

STREAM = MovieActor ⋈ MoviePlatform ⋈ ActorPlatform

Result

The decomposition removes redundancy caused by join dependencies and gives a 5NF (Project-Join Normal Form) design with lossless reconstruction.